

Forecasting Chaos Across Model Families: A Controlled Benchmark of LLMs, Time-Series Foundation Models, Learned Dynamics, and Numerical Integrators on k -Pendulum Systems

Sriman Narayan Kandi*, Trishant Srinivasan, Shrithik Shahapure
kandisriman@gmail.com

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Abstract

We ask a deliberately operational question: given the state of a chaotic mechanical system at time t , which family of modern models best predicts its state at $t + T$, and at what reliability and cost? We build a controlled benchmark on k -link pendulums ($k \in \{1, 2, 3\}$; chaotic for $k \geq 2$) that places four model families on identical inputs and metrics: frontier large language models (LLMs), a time-series foundation model (Chronos), learned dynamics models (Neural ODE, Hamiltonian and Lagrangian neural networks), and classical integrators (Euler, RK4, symplectic). Crucially, every family is evaluated on the *same shared out-of-sample held-out trajectories*, so the learned models are not scored on their own training data—a confound that, we show, inflates them by roughly $3\times$. Three contributions follow. **(i) A reliability-adjusted evaluation.** Averaging error only over a model’s *answered* cells rewards selective abstention; we instead score unanswered or invalid cells at a random-guess baseline, which reorders the leaderboard and exposes a brittle frontier LLM that answers only 82% of cells. **(ii) Out-of-sample, the ranking inverts.** On the shared held-out set, classical integrators dominate, and the best *non-numerical* forecaster is a frontier LLM (kimi, 0.538 rad), ahead of the time-series model and *every* learned model—which collapse from an in-sample ~ 0.26 to 0.84–1.06 rad once tested off their training trajectories. The training objective still matters (rollout loss beats derivative matching, $p_{\text{Holm}} = 1.6 \times 10^{-7}$), but the in-sample “mixed- k wins” result does not survive: mixed- k is statistically indistinguishable from single- k . **(iii) Chaos-appropriate analysis.** A predictability-horizon metric shows only numerical integrators track the system below strict error thresholds past 1 s. A confound-free, matched-constant system-identification test then dismantles an apparent “LLMs do better with hidden constants” effect: once initial conditions are held fixed, disclosing the constants is no worse (and at $k=1$ clearly better), and a direct inference probe shows the LLMs do not recover the constants—they fall back on Earth-standard priors. Overall, for chaotic forecasting the headline is a cautionary one: in-sample evaluation of learned dynamics models is badly optimistic, and on a level out-of-sample field a frontier LLM is the strongest learned forecaster while classical numerics remain unbeaten.

1 Introduction

Large language models are increasingly applied to scientific tasks, and time-series foundation models now offer general-purpose forecasting out of the box. A natural question for physical modelling is whether either family can serve as a *simulator*: given the state of a dynamical system at time t , can it predict the state at $t + T$? The k -link pendulum is an ideal probe. Single pendulums ($k=1$) are integrable; double and triple pendulums ($k \geq 2$) are chaotic with a positive Lyapunov exponent. The task is unambiguous (predict angles and angular velocities), ground truth is computable to arbitrary precision, and grading requires no domain expertise.

Prior evaluations tend to study one family in isolation: LLMs on physics word problems, time-series foundation models on forecasting suites, or physics-informed networks on chaotic benchmarks. We instead place all four

*Corresponding author: kandisriman@gmail.com

families on a single mechanical substrate under *identical* inputs, horizons, and metrics, and we take reliability and the structure of chaos seriously as first-class evaluation concerns rather than footnotes.

Contributions.

1. A unified, reproducible cross-family benchmark (LLMs, a time-series foundation model, learned dynamics, and integrators) on k -pendulum forecasting, in which all families are scored on the *same out-of-sample held-out trajectories* and a **reliability-adjusted metric** scores abstentions instead of discarding them (§3.3, §5.1).
2. An **out-of-sample leaderboard that inverts the in-sample story**: tested off their training trajectories the learned models collapse ($\sim 0.26 \rightarrow 0.84\text{--}1.06$ rad), and the best non-numerical forecaster is a frontier LLM (kimi, 0.538 rad). The training objective still matters—rollout loss beats derivative matching significantly—but mixed- k rollout, the in-sample winner, is a statistical null out-of-sample (§5.3).
3. **Chaos-appropriate analysis**: a predictability-horizon metric (only numerical integrators track past 1 s at strict thresholds) and a *confound-free* system-identification test. A naive regime contrast suggests LLMs do better with constants hidden; a matched-constant paired experiment plus a direct inference probe shows this is a difficulty confound—the LLMs do not infer the constants but fall back on Earth-standard priors (§5.4).

2 Related Work

Learned dynamics. Neural ODEs learn a continuous-time vector field from data [Chen et al., 2018]. Hamiltonian Neural Networks impose an energy-conserving inductive bias by deriving dynamics from a learned Hamiltonian [Greydanus et al., 2019]; Lagrangian Neural Networks enforce the Euler–Lagrange equations through second-order autodifferentiation [Cranmer et al., 2020]. Chaotic systems such as the Lorenz attractor and the double pendulum are standard testbeds for such models [Gilpin, 2021]. We benchmark all three against general-purpose models under matched conditions.

Foundation models as forecasters. LLMs can act as zero-shot time-series forecasters when series are serialised as text [Gruver et al., 2023], and dedicated time-series foundation models—Chronos [Ansari et al., 2024], TimesFM [Das et al., 2024], and Moirai [Woo et al., 2024]—are trained for general forecasting. Zhang and Gilpin [2025] study zero-shot forecasting of chaotic systems with such models. Our benchmark adds these forecasters to the same grid as physics-informed and language models.

LLMs for physical and mathematical reasoning. LLMs show strong results on mathematical problem solving [Lewkowycz et al., 2022] and, with chain-of-thought prompting, on multi-step reasoning [Wei et al., 2022]. Recent benchmarks probe physics reasoning in LLMs more directly [Xu et al., 2025]. Our task differs: it demands quantitative trajectory prediction of a chaotic system, not qualitative or symbolic reasoning.

3 Benchmark Design

3.1 Task and systems

Given the state $\mathbf{s}_0 = (\theta_1, \dots, \theta_k, \omega_1, \dots, \omega_k) \in \mathbb{R}^{2k}$ of a k -link pendulum at $t=0$, predict $\mathbf{s}_T$ at a horizon T . Ground-truth trajectories use the full nonlinear Lagrangian equations of motion integrated with `scipy`’s DOP853 at $\Delta t=10^{-3}$ s (rtol 10^{-10} , atol 10^{-12}). Initial conditions are sampled $\theta_i \sim \mathcal{U}(-3, 3)$ rad, $\omega_i \sim \mathcal{U}(-1, 1)$ rad/s with a fixed seed.

3.2 Regimes and grid

Three regimes test generalisation to non-standard constants (Table 1). In `changed_hidden` the constants (g, L, m , damping) are *not* disclosed to the model, turning the cell into an in-context system-identification

Table 1: Physical regimes. `changed_hidden` withholds the constants.

Regime	g (m/s ²)	L (m)	m (kg)	damping	disclosed
<code>normal</code>	9.81	1.0	1.0	0.0	—
<code>changed_disclosed</code>	3.71	1.4/0.7/1.1	2.0/0.6/1.5	0.05	yes
<code>changed_hidden</code>	1.62	0.8/1.2/0.9	0.5/1.8/1.0	0.10	no

problem. An evaluation *cell* is a tuple (model, k , regime, modality, horizon, prompting); horizons are $\{0.01, 1, 10, 60\}$ s, modalities $\{\text{coords}, \text{image}, \text{image+coords}\}$ (LLMs only), and prompting $\{\text{direct}, \text{chain-of-thought}\}$ (LLMs only). Non-LLM models collapse to the coordinate / direct setting.

Out-of-sample protocol. The central methodological point of this paper is that learned dynamics models must be tested off their training data. We therefore generate a *shared held-out set* (a separate seed from the training trajectories) of 180 initial conditions (60 per regime, spanning $k \in \{1, 2, 3\}$) and evaluate *every* family on the identical trajectories, keyed by `movement_id`. The cross-family leaderboard (§5.1) is computed on the discriminating horizons $\{1, 10\}$ s, where chaos has begun to bite but error has not yet saturated (§3.3), giving $180 \times 2 = 360$ comparison cells per model. Earlier in-sample numbers—where the learned models were scored on their own training trajectories—are reported nowhere in this version; we quantify the size of that leak directly (§5.1).

3.3 Metrics

The primary error is the mean absolute angle error (radians), wrapped to $[-\pi, \pi]$ and averaged over links and trajectories; we also track Cartesian bob error, angular-velocity sign match, and energy drift. Three evaluation choices are specific to this setting.

Reliability-adjusted error. A cell is *answered* only if the model returns a parseable, finite prediction. Reporting error over answered cells alone rewards a model for abstaining on hard cells. We therefore report a reliability-adjusted error in which unanswered cells are scored at the random-guess baseline $\pi/2$ (the expected wrapped-angle error of a uniformly random guess); we also report the strict upper bound π . This makes *answering* part of the score, not a separate column to be overlooked.

Predictability horizon. For chaotic systems, pointwise error past the Lyapunov time saturates to a system-dependent constant for *every* model, so long-horizon error rankings are weakly informative. We define the predictability horizon as the longest horizon a model tracks *contiguously* below a threshold τ , starting from $t=0$: we walk horizons in increasing order and stop at the first that exceeds τ . The contiguous rule is deliberate—once a trajectory has diverged, a later horizon can dip back below τ by angle-wrap coincidence, which is not predictability. We sweep τ to expose threshold sensitivity.

System-identification gap. The gap $\Delta_{\text{sysid}} = \text{err}(\text{hidden}) - \text{err}(\text{disclosed})$ isolates the cost of *inferring* the constants versus being told them. A model that merely substitutes given numbers is hurt by hiding them ($\Delta > 0$); a model that infers them in context is not ($\Delta \leq 0$). Measuring this cleanly requires holding the trajectory fixed and toggling *only* disclosure—otherwise disclosure is confounded with the (different) constants and initial conditions of each regime. We therefore run a dedicated matched experiment (§5.4): 5 non-standard constant sets, 10 trajectories each, $k \in \{1, 2, 3\}$, evaluated under disclosed and hidden conditions *on identical trajectories*, with a direct probe that asks the hidden-condition model to report its inferred constants.

4 Methods

LLMs. We evaluate `kimi-k2.6`, `grok-4-1-fast-reasoning`, and `deepseek-v4-pro` through an OpenAI-compatible API, with a structured prompt (system description, disclosed constants where applicable, initial

state) and a JSON answer schema (Appendix A). Reasoning models that emit long chains-of-thought require large token budgets and timeouts, so we use a uniform reliability-fixed configuration (65,536 max tokens, 300s timeout) for all three; reasoning traces are parsed from the dedicated reasoning channel when present. We take a *single* deterministic sample per cell (temperature 0); no best-of- N aggregation is used, so reliability reflects the raw per-call behaviour. The total API cost of the out-of-sample LLM evaluation (Parts A and B, 3,960 cells) was \$110.

Time-series foundation models. We run Chronos [Ansari et al., 2024] locally on a single GPU via a lightweight adapter that resamples the pre- $t=0$ history onto a uniform grid and forecasts the $2k$ state channels. The out-of-sample leaderboard uses **chronos-2**, which forecasts either each channel independently (*univariate*) or all $2k$ channels jointly (*multivariate*); this lets us test, with one model, whether modelling cross-channel coupling helps on a coupled dynamical system. For that test we run a dedicated study with 20 trajectories per (k, regime) cell and compare the two modes *paired* on identical trajectories (Wilcoxon signed-rank with bootstrap confidence intervals and Holm correction). (The earlier univariate **chronos-t5-large** served only in a preliminary round; TimesFM is univariate and adds no multivariate signal; Moirai’s **uni2ts** could not be installed in our environment—all are left to future rounds.)

Learned dynamics. Neural ODE, HNN, and LNN share a 256-wide, 3-layer tanh MLP and integrate at inference with **scipy**. We compare two objectives: *derivative matching* (MSE on finite-difference $\dot{\mathbf{s}}$) and *rollout loss* (MSE on the state after k Euler steps, with gradients flowing through the unroll). We introduce *mixed- k rollout*, which trains on several window lengths jointly (Neural ODE: $k \in \{10, 50\}$; HNN: $k \in \{5, 20\}$), optimising short, medium, and long horizons at once. The LNN’s loss needs second-order autodifferentiation, which Apple’s MPS backend cannot run; we train it at full scale (67k samples, hidden 256) on a CUDA GPU—the configuration its prior incarnation could not reach.

Integrators. Euler, RK4, and Störmer–Verlet (symplectic) at $\Delta t=0.01$ s using the exact equations of motion; their error is purely from discretisation. RK4 is the data-generating process and therefore an oracle ceiling, not a competitor.

5 Results

5.1 Out-of-sample leaderboard

Table 2 ranks all 15 models on the shared out-of-sample held-out set (360 cells: 180 trajectories $\times$ horizons $\{1, 10\}$ s), using the reliability-adjusted metric so that an unanswered or missing cell is scored at the random-guess baseline $\pi/2$ rather than dropped. Every model is scored on the *identical* grid; **n_missing** counts cells a model never produced (only the LNN, whose autograd-ODE integration diverges to NaN on 55% of cells, is affected). LLMs are scored on their direct-prompt forecasts, the setting all families share. Confidence intervals are 95% paired bootstrap ($B=10^4$).

Three things stand out. First, classical integrators dominate: RK4 (the data-generating oracle) at 0.059, symplectic at 0.205, Euler at 0.405. Second, **the best non-numerical forecaster is a frontier LLM—kimi** at 0.538 rad (99% answered)—ahead of both Chronos modes and, strikingly, *every* learned dynamics model. Third, the learned models, which in an in-sample evaluation scored ~ 0.26 rad, collapse to 0.84–1.06 rad once they are tested on trajectories they were not trained on: the best of them (mixed- k rollout Neural ODE, 0.844) ranks 6th, behind the integrators, a zero-shot LLM (**kimi**), and both zero-shot Chronos modes. This $\sim 3\times$ gap between in-sample and out-of-sample error is the single most important number in the paper: it means the widely-reported strength of small learned dynamics models on this benchmark was substantially an artefact of evaluating them on their own training data.

Reliability is load-bearing. The success column is not a footnote. **grok** answers only 82% of the held-out grid and **lln** only 45%; scoring those abstentions at $\pi/2$ (rather than averaging over answered cells alone, which would reward them for skipping hard cells) is what places them at ranks 8 and 14. A naive answered-only

Table 2: Out-of-sample leaderboard on the shared held-out set (360 cells, horizons $\{1, 10\}$ s). Reliability-adjusted mean angle error (rad): unanswered or missing cells scored at the random-guess baseline $\pi/2$; 95% bootstrap CI. RK4 is the data-generating oracle. New in this work marked $\dagger$.

#	Model	Err	95% CI	Answered	Family
–	rk4 (oracle)	0.059	[0.032, 0.089]	100%	numerical
1	symplectic	0.205	[0.163, 0.251]	100%	numerical
2	euler	0.405	[0.344, 0.470]	100%	numerical
3	kimi-k2.6	0.538	[0.475, 0.602]	99%	LLM
4	chronos2-multi $\dagger$	0.684	[0.617, 0.752]	100%	time-series
5	chronos2-uni $\dagger$	0.697	[0.631, 0.766]	100%	time-series
6	neural-ode-rollout-mixed $\dagger$	0.844	[0.770, 0.920]	100%	learned
7	neural-ode-rollout	0.870	[0.795, 0.947]	100%	learned
8	grok-4-1-fast-reasoning	0.881	[0.805, 0.957]	82%	LLM
9	hnn-rollout	0.922	[0.847, 0.998]	100%	learned
10	deepseek-v4-pro	0.952	[0.881, 1.027]	100%	LLM
11	hnn-rollout-mixed $\dagger$	1.003	[0.923, 1.084]	100%	learned
12	neural-ode	1.028	[0.953, 1.107]	100%	learned
13	hnn	1.064	[0.982, 1.145]	100%	learned
14	lnn $\dagger$ (full-scale)	1.421	[1.363, 1.481]	45%	learned

Table 3: Out-of-sample mean angle error (rad) by horizon, reliability-adjusted, on the shared held-out grid (selected models). The cross-family comparison uses the discriminating horizons $\{1, 10\}$ s.

Model	1 s	10 s
rk4 (oracle)	0.000	0.118
symplectic	0.042	0.368
euler	0.067	0.743
kimi-k2.6	0.457	0.619
chronos2-multi	0.591	0.777
chronos2-uni	0.622	0.773
neural-ode-rollout-mixed	0.779	0.908
neural-ode-rollout	0.790	0.950
hnn-rollout	0.758	1.085
grok-4-1-fast-reasoning	0.905	0.857
neural-ode	0.793	1.264
deepseek-v4-pro	0.996	0.909
lnn (full-scale)	1.335	1.508

leaderboard would flatter exactly the least reliable models. We regard answering as part of the score, not a separate column to be overlooked.

5.2 Behaviour by horizon

Table 3 splits the out-of-sample error by horizon. At 1 s the ordering already mirrors the leaderboard: the integrators are near-exact, **kimi** (0.457) is the best non-numerical model, and the learned models sit at 0.76–0.97. At 10 s every model is well into the chaotic regime, but the ranking is stable rather than scrambled—**kimi** (0.619) still leads the non-numerical field while the learned models drift to 0.91–1.26. Notably the mixed- k Neural ODE, which the in-sample evaluation singled out as the standout at 1–10 s, is here merely mid-pack (0.779 / 0.908), worse than a zero-shot LLM at both horizons. The qualitative LLM signature from earlier rounds—confident near $t=0$, regressing to a generic chaotic guess later—is consistent with these numbers but is no longer a disqualifier relative to the (now equally struggling) learned models.

Table 4: Training-objective study, out-of-sample mean angle error (rad) on the shared held-out grid (horizons $\{1, 10\}$ s); architecture fixed, only the loss/window changes. Significance from a dedicated powered paired study on the full four-horizon held-out set ($n=720$, Wilcoxon+Holm).

Model	Objective	Err	vs. prev. row
Neural ODE	derivative MSE	1.028	—
	rollout (single- k)	0.870	sig. ($p_{\text{Holm}}=1.6\times 10^{-7}$)
	mixed- k	0.844	n.s. ($p_{\text{Holm}}=0.089$)
HNN	derivative MSE	1.064	—
	rollout (single- k)	0.922	(rollout helps)
	mixed- k	1.003	n.s. ($p_{\text{Holm}}=0.088$)

5.3 The training objective matters; mixed- k does not

Table 4 isolates the effect of the loss while holding the architecture fixed, on the out-of-sample grid. One effect is robust: **rollout loss decisively beats derivative matching**. For the Neural ODE, derivative matching (1.028) drops to 0.870 under a single- k rollout loss; in a dedicated powered paired study ($n=720$ trajectories, Wilcoxon signed-rank) this gap is highly significant ($p_{\text{Holm}}=1.6\times 10^{-7}$). Penalising trajectory error through an unrolled integrator, rather than instantaneous derivative error, directly optimises what the benchmark measures.

The in-sample story does *not* survive contact with out-of-sample data, however. **Mixed- k rollout—training on several window lengths jointly, the previously reported headline winner—is a statistical null**. For the Neural ODE it is indistinguishable from single- k (0.844 vs. 0.870; powered $p_{\text{Holm}}=0.089$), and for the HNN it is numerically *worse* (1.003 vs. 0.922; $p_{\text{Holm}}=0.088$). What looked like an architecture-agnostic benefit of mixing horizons was, on held-out trajectories, within noise. Finally, full-scale CUDA training of the LNN—finally running the second-order autograd its loss requires—leaves it the *worst* model (1.421), with 55% of cells diverging to NaN under autograd-ODE integration; its weakness is intrinsic to the Lagrangian-net+ODE formulation on this chaotic system, not a hardware artefact.

5.4 Chaos-appropriate analysis

Predictability horizon. Table 5 reports the longest horizon each model tracks *contiguously* below a threshold τ , over the full horizon ladder $\{0.01, 1, 10, 60\}$ s available for the non-LLM models. Out-of-sample, **only the numerical integrators stay below strict thresholds past 1 s**: RK4 tracks to 10 s at $\tau=0.25$ and to 60 s at $\tau=0.5$, symplectic to 60 s at $\tau=0.5$, Euler to 60 s only at the loose $\tau=1.0$. Every learned model—including the rollout and mixed- k Neural ODEs that the in-sample analysis credited with tracking to 10 s—collapses to the first rung (0.01 s) at all strict thresholds and only “reaches” 60 s at $\tau=1.0$ ($\sim 57^\circ$), where the metric is meaningless. The LLMs were evaluated only at $\{1, 10\}$ s on the shared set, so their ladder is coarser; within it, none tracks below 0.5 rad at 1 s except *kimi*. The sweep underlines the leaderboard: on held-out trajectories no data-driven model has a usable predictability horizon beyond what numerics already provide.

A confounded “hidden helps” effect... On the shared held-out set the three regimes differ in both their constants *and* their initial conditions, so a regime contrast conflates disclosure with difficulty. Taken at face value that contrast is striking: pooled across LLMs, the hidden regime has *lower* error than the disclosed regime (-0.390 rad, $p_{\text{Holm}}<10^{-3}$, unpaired) and than the normal regime (-0.524 , $p_{\text{Holm}}<10^{-3}$). Read naively, this is the “LLMs do better when you hide the constants” result reported in earlier rounds, and it is tempting to call it in-context system identification.

... that a matched experiment dismantles. We therefore ran the confound-free test of §3.3: 5 non-standard constant sets, 10 trajectories each, $k \in \{1, 2, 3\}$, scored under disclosed and hidden conditions on *identical* trajectories, paired by trajectory (Table 6). The effect reverses. Overall, hiding the constants makes predictions slightly *worse*, not better ($\Delta_{\text{sysid}}=+0.069$ rad, $p_{\text{Holm}}=0.034$); at $k=1$ disclosure helps strongly

Table 5: Predictability horizon (s): longest horizon tracked contiguously below threshold τ , on the shared out-of-sample set (full ladder $\{0.01, 1, 10, 60\}$ s). Higher is better.

Model	$\tau=0.1$	0.25	0.5	1.0
rk4 (oracle)	1.0	10	60	60
symplectic	1.0	1.0	60	60
euler	1.0	1.0	1.0	60
neural-ode-rollout	0.01	0.01	0.01	60
neural-ode-rollout-mixed	0.01	0.01	0.01	60
chronos2-multi	0.01	0.01	0.01	60
chronos2-uni	0.01	0.01	0.01	60
hnn-rollout	0.01	0.01	0.01	1.0
neural-ode	0.01	0.01	0.01	1.0
hnn	0.01	0.01	0.01	1.0
lnn	0.01	0.01	0.01	0.01

Table 6: Matched, confound-free system-identification gap (rad), paired on identical trajectories. $\Delta_{\text{sysid}} = \text{err}(\text{hidden}) - \text{err}(\text{disclosed})$; *positive* means the model does *worse* without being told the constants. Reliability-adjusted; paired Wilcoxon, Holm-corrected.

Model	disclosed	hidden	Δ_{sysid}
kimi-k2.6	0.746	0.957	+0.211
grok-4-1-fast-reasoning	0.987	1.025	+0.038
deepseek-v4-pro	1.425	1.382	−0.042
pooled (paired, $n=900$)			+0.069*
$k=1$			+0.297***
$k=3$			−0.055*

* $p_{\text{Holm}} < 0.05$, *** $p_{\text{Holm}} < 10^{-3}$.

(+0.297, $p_{\text{Holm}} < 10^{-3}$) and at the 1 s horizon as well (+0.145, $p_{\text{Holm}} < 10^{-3}$), with only a small reverse effect at $k=3$ (−0.055, $p_{\text{Holm}} = 0.016$). Per model, **kimi** is clearly hurt by hiding ($\Delta = +0.211$), **grok** mildly (+0.038), and **deepseek** is flat (−0.042). The apparent advantage of the hidden regime in the unmatched contrast was thus a difficulty confound—the hidden-regime trajectories were simply easier—not evidence of inference.

The inference probe confirms it. In the hidden condition we additionally asked each model to report its inferred constants. The models do *not* recover them: against true $g \in [1.6, 12]$ the inferred- g MAE is 1.81 (**deepseek**), 1.88 (**grok**) and 2.12 (**kimi**). The failure mode is uniform—they fall back on Earth-standard priors: **deepseek** answers $g \approx 9.81$ on essentially every cell, and **grok** and **kimi** answer $L=m=1.0$ on essentially every cell (a constant output, so the inferred-vs-true rank correlation is undefined). The one weak signal is **deepseek**’s rod lengths, which correlate with truth at Spearman 0.11 ($p=0.007$). There is, in short, little evidence that any accuracy difference between regimes is *caused* by in-context system identification; the earlier interpretation does not hold up once the confound is removed and the mechanism is probed directly.

5.5 Does multivariate modelling help? A powered null

The headline benchmark uses one trajectory per cell, which is adequate for the large gaps between families but too thin to resolve small effects. To test the natural hypothesis that a time-series model able to share information across the $2k$ state channels should benefit from their coupling, we run a dedicated powered study: Chronos-2 in univariate and multivariate modes, 20 trajectories per (k , regime) cell ($n=720$ paired samples), compared paired on identical trajectories.

The hypothesis does *not* hold. The two modes are statistically indistinguishable in practice: mean error 0.549 (uni, 95 % CI [0.505, 0.593]) vs. 0.543 (multi, [0.498, 0.589]). The paired difference is negligible (+0.005 rad overall) and shows *no* significant effect at any k or at the discriminating horizons (1 s, 10 s); the per- k trend

seen at $n=1$ (multivariate appearing to help most at $k=3$) *reverses* to zero at $n=720$, confirming it was sampling noise. A nominal overall $p=0.025$ is driven entirely by a sub-0.01 rad bias at the trivial 0.01 s horizon and does not survive Holm correction. We therefore report a clean null: **cross-channel coupling gives no practically meaningful benefit for time-series forecasting of this coupled chaotic system**. (On the shared out-of-sample grid the Chronos family lands at 0.68–0.70 rad—ahead of *every* learned model and behind only the numerical integrators and *kimi* in Table 2.) This result is a concrete illustration of why error bars matter here: an eye-catching $n=1$ trend became a null under proper sampling.

6 Discussion

In-sample evaluation of learned dynamics is badly optimistic. The single most consequential result is methodological: when the learned dynamics models are scored on the *same* held-out trajectories as everyone else, rather than on their own training data, their error roughly triples ($\sim 0.26 \rightarrow 0.84\text{--}1.06$ rad) and they fall *below* a zero-shot time-series model and a zero-shot LLM. The “small fitted model beats general-purpose forecasters” conclusion that an in-sample leaderboard invites does not survive a level playing field. For chaotic systems the lesson is concrete: a learned simulator must be reported on trajectories it never saw, or its headline number is largely a memorisation artefact.

On a level field, a frontier LLM is the strongest learned forecaster. Out of sample the best non-numerical model is *kimi* (0.538 rad), ahead of both Chronos modes (0.68–0.70) and every learned dynamics model (0.84–1.06); only the classical integrators remain clearly ahead. The training *objective* still moves the needle—rollout loss decisively beats derivative matching—but the in-sample “mixed- k wins” headline is a statistical null on held-out data. What generalises is the integrator-shaped loss, not the particular window-mixing recipe.

Reliability is a first-class axis. The naive answered-only leaderboard flatters exactly the least reliable models. Scoring abstentions at the random-guess baseline is what places *grok* (answers only 82% of the held-out grid) at rank 8 and the LNN (45%) last. Chain-of-thought does not rescue this either: paired CoT-vs-direct differences are statistically insignificant for every LLM (the largest model, *kimi*, is if anything worse with CoT). Numerical integration is inherently sequential, and a fixed budget of reasoning tokens cannot stand in for thousands of integration steps. Any benchmark that reports error over answered cells alone is measuring the wrong thing for deployment.

No evidence of in-context system identification. The apparent “LLMs do better when constants are hidden” effect—which an in-sample reading took as a bright spot worth scaling—is a difficulty confound. In a matched experiment that toggles *only* disclosure on identical trajectories, hiding the constants is if anything slightly *worse* ($+0.069$ rad overall; $+0.297$ at $k=1$), and a direct inference probe shows the models do not recover the constants at all: they fall back on Earth-standard priors (g MAE $\sim 1.8\text{--}2.1$ against true $g \in [1.6, 12]$). The one capability the optimistic story singled out does not, on inspection, exist; we caution against reading regime contrasts as inference without a confound-free matched test.

7 Limitations

The out-of-sample leaderboard aggregates 180 held-out trajectories (60 per regime) with bootstrap confidence intervals, and the powered sub-studies (objective, multivariate, system-ID) use $n=720\text{--}900$ paired samples; still, all trajectories derive from a single held-out seed, and additional seeds would further tighten the headline error bars. Predictions are single-endpoint rather than full-trajectory; a full-rollout protocol would let the predictability horizon be measured directly rather than inferred from four horizons. The comparison remains structurally asymmetric: the learned models are trained on the k -pendulum family and then tested on *held-out* trajectories of it, whereas the LLMs and Chronos are zero-shot—fair on the shared grid, but not a like-for-like training regime. The pendulum equations are ubiquitous in training corpora, so contamination cannot be excluded; the hidden-constants regime partially mitigates this but does not eliminate it. Finally, the study

is a single mechanical substrate and three LLMs; generality across other chaotic systems (Lorenz, driven pendulum) and a broader model roster (e.g. Qwen-VL, TimesFM, Moirai) is future work.

8 Conclusion

On a controlled k -pendulum benchmark spanning four model families under identical conditions and—crucially—on the *same out-of-sample held-out trajectories*, the headline is cautionary. In-sample evaluation of learned dynamics models is badly optimistic: scored off their training data the learned models triple their error and drop behind both a zero-shot time-series model and a frontier LLM, so that on a level field `kimi` is the best non-numerical forecaster and classical numerics remain unbeaten. The training objective, not the architecture, is what generalises—rollout loss beats derivative matching—but the in-sample “mixed- k wins” result is a statistical null out of sample. Reliability-adjusted scoring and chaos-appropriate metrics overturn the naive leaderboard, and a confound-free matched experiment plus a direct inference probe dismantle the one apparent bright spot for general models: there is no in-context system identification, only Earth-prior anchoring. The lesson worth carrying forward is methodological—test learned simulators out of sample, score abstentions, and demand a matched control before crediting inference.

Reproducibility. Code, prompts, trained weights, and per-cell checkpoints are released with the benchmark. All training and evaluation in this round run on a single consumer GPU (8GB); no proprietary training data is used.

Author Contributions

S.N.K. (lead author, corresponding) conceived and designed the benchmark, built the evaluation harness, the learned-dynamics and time-series components, and the analysis pipeline, ran the out-of-sample and significance experiments, and wrote the manuscript. T.S. contributed to designing the benchmark and building the evaluation harness, and the learned-dynamics and time-series components. S.S. contributed to running and evaluating the frontier LLM experiments. Author order reflects relative contribution; S.N.K. is the corresponding author. All authors reviewed and approved the final manuscript.

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A Prompt templates

The coordinate, direct-prompt template provides the system description, the disclosed constants (omitted in `changed_hidden`), and the initial state, and requires a JSON object `{"theta": [...], "omega": [...]}`. The chain-of-thought variant prepends a step-by-step reasoning instruction. Full templates are in the released code.

B Training and compute

All learned models use a 256-wide, 3-layer tanh MLP, Adam, cosine-annealed learning rate, on a single NVIDIA GTX 1070 Ti (8 GB). Mixed- k rollout adds negligible parameters; its cost is wall-clock (the HNN mixed- k run is the most expensive at ~ 6.7 h because each unroll step recomputes ∇H via autograd). Chronos runs at inference only. Dataset: 67,491 samples across $k \in \{1, 2, 3\}$ and three regimes, subsampled at $\Delta t = 0.01$ s from the 10^{-3} s ground truth.